

A Search For Rapid Rotation In Massive White Dwarf Stars

ETAMU Spring 2025 Annual Research Symposium

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Advisor: Doctor Kurtis Williams
University: ETAMU



EAST TEXAS A&M
— UNIVERSITY —

Stellar Evolution

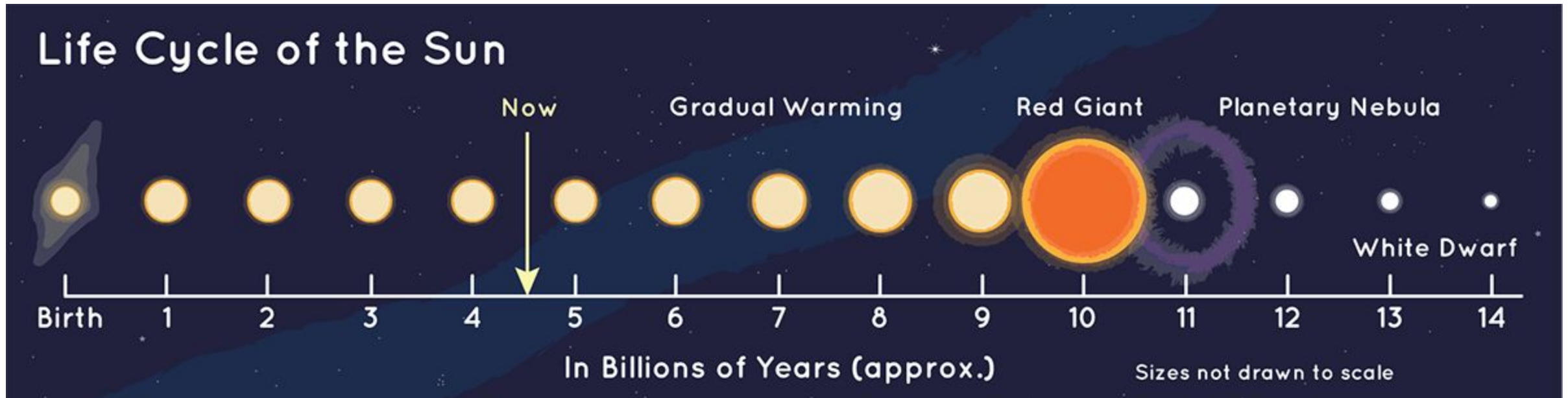
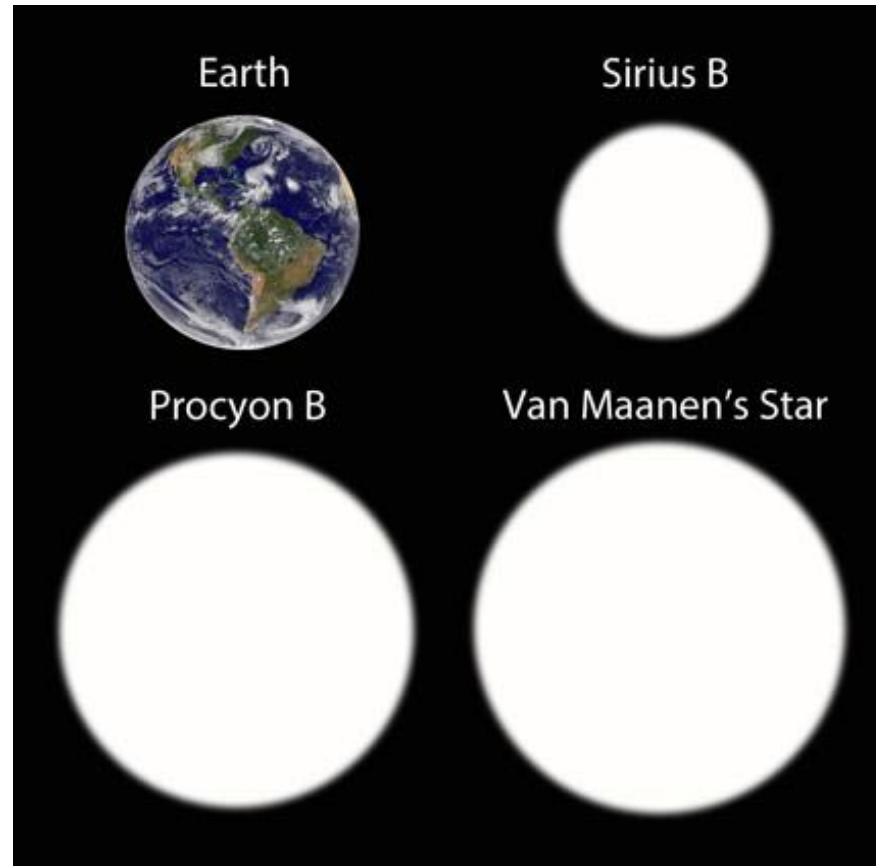


Image Credit: NASA; <https://spaceplace.nasa.gov>



White Dwarf Stars (WDs)



My Project

Goal: Search for variation and rapid rotation in massive WDs.

- Why might it vary in luminosity?
 - Starspots
- Why might it rapidly rotate?
 - Massive Sun-like star
 - Double Degenerate Merger
- What is a rapidly rotating massive WD?
 - Period ≤ 6 hours
 - Mass $\leq 0.9 M_{\odot}$

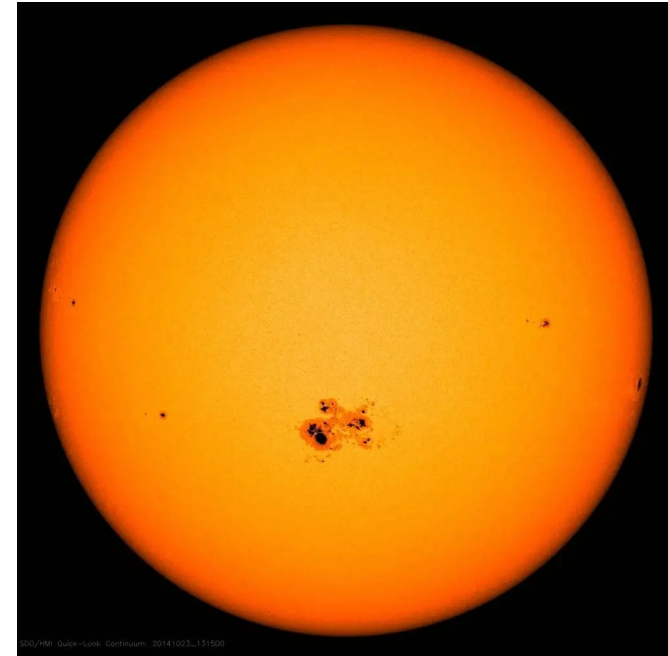


Image credit: NASA/SDO



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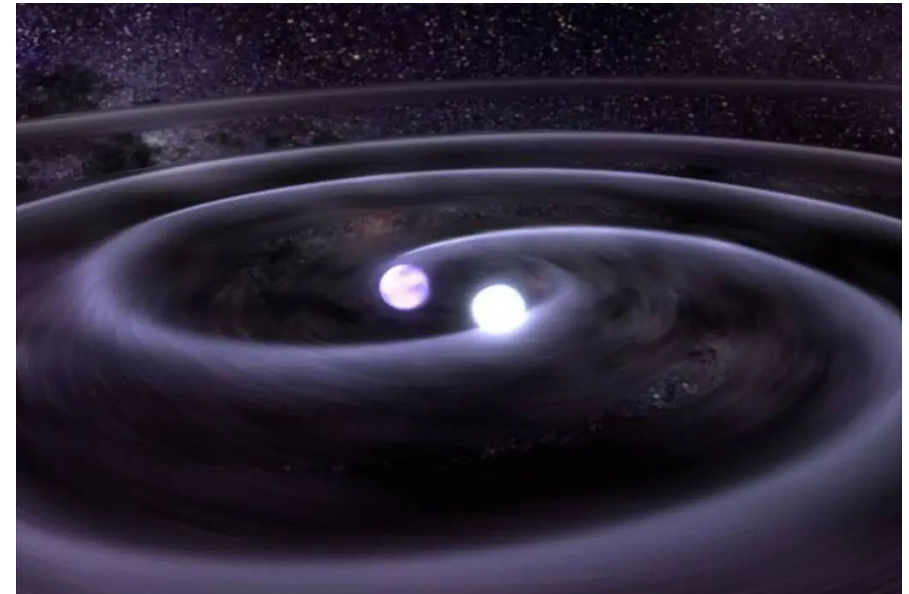
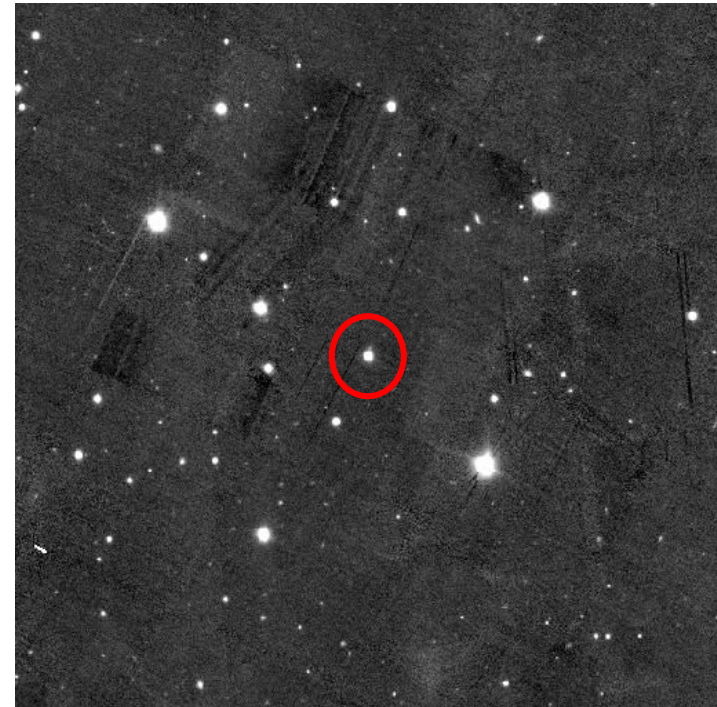
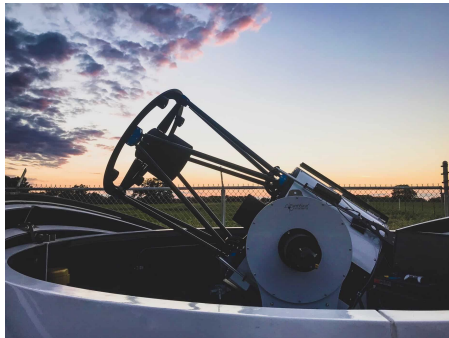


Image Credit: NASA/Tod Strohmayer (GSFC)/Dana Berry (Chandra X-Ray Observatory)



Target Observed

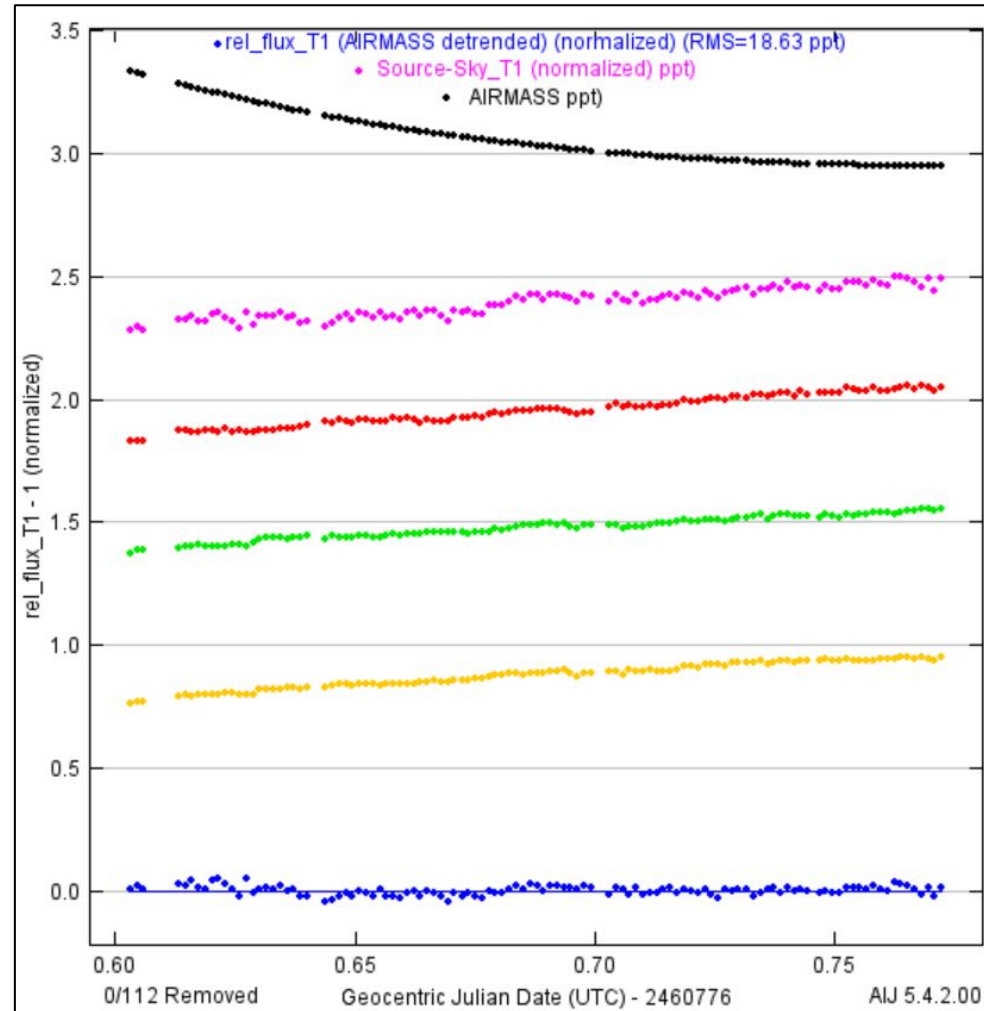
- LAMOST J132743.78+545725.7
 - $M \approx 1.05 M_{\odot}$
 - $G = 14.60$
 - Observed 5 days (2 sessions)
 - UT July 2nd-4th, 2024
 - UT April 11th-12th, 2025



Info: MWDD, Image Credit: ETAMU and PanSTARRS-1

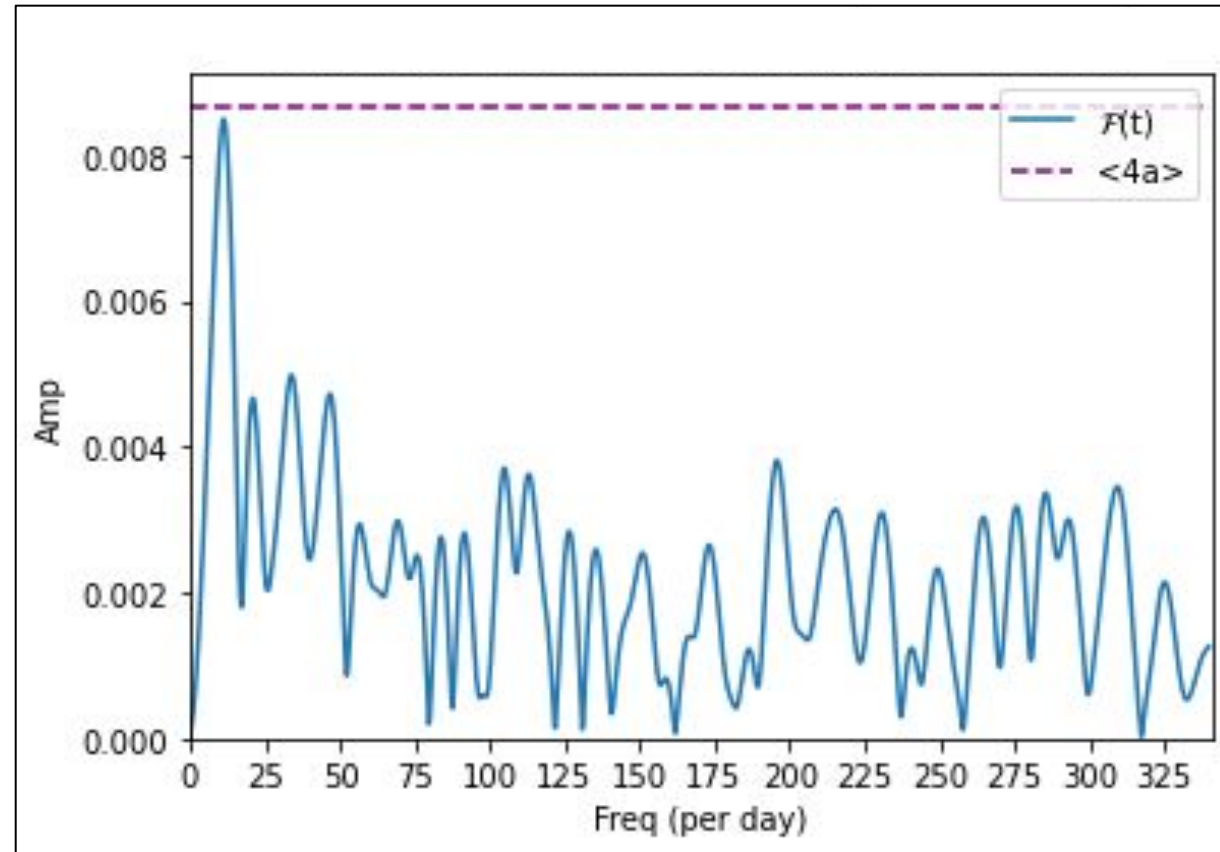


Photometry - UT April 11th, 2025



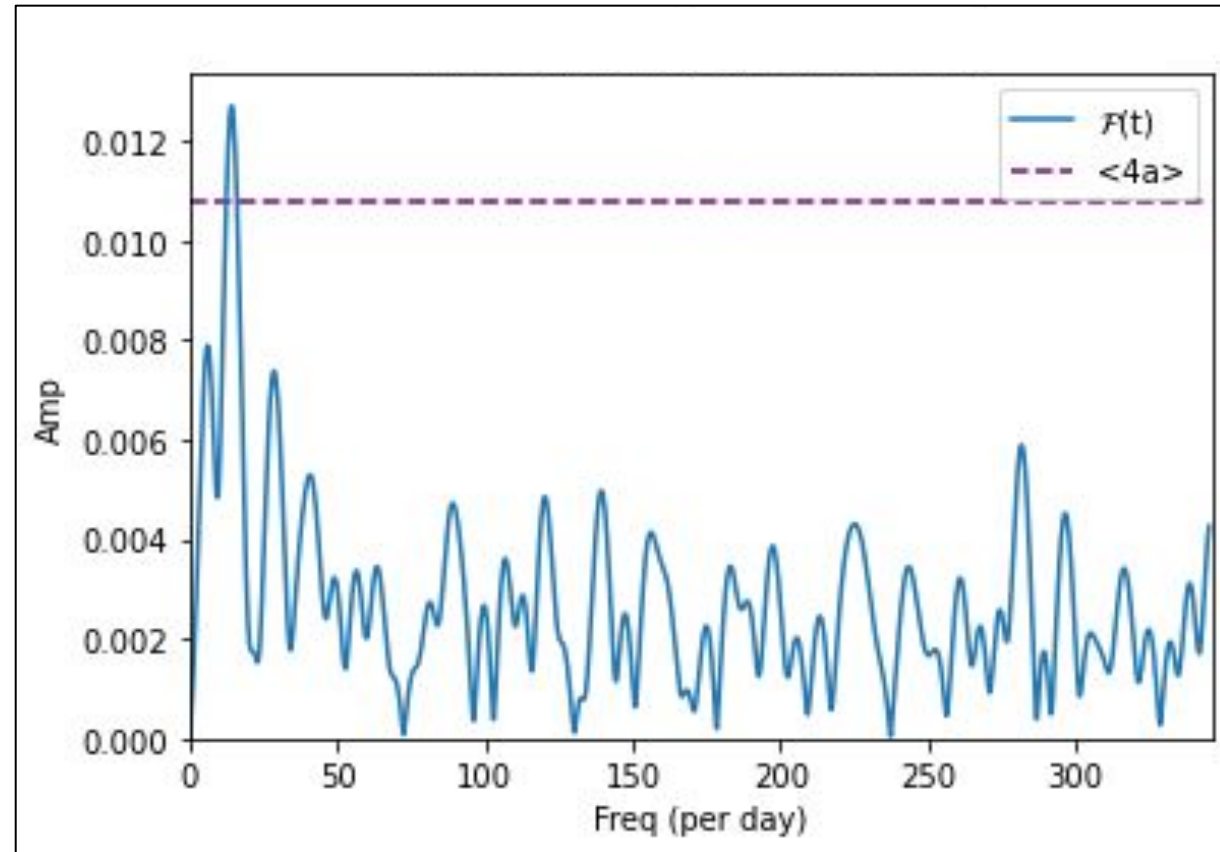
Periodogram - UT July 3rd, 2024

- Freq ≈ 11.48 /day
- Period ≈ 2.09 hours
- Amp ≈ 0.008



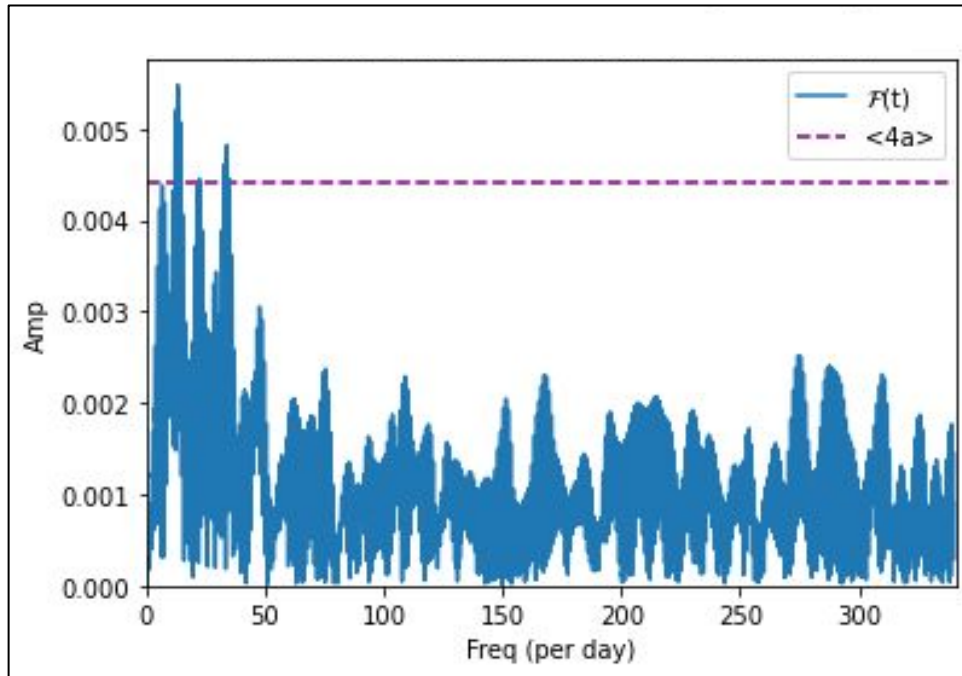
Periodogram - UT April 11th, 2025

- Freq ≈ 14.20 /day
- Period ≈ 1.69 hours
- Amp ≈ 0.0127

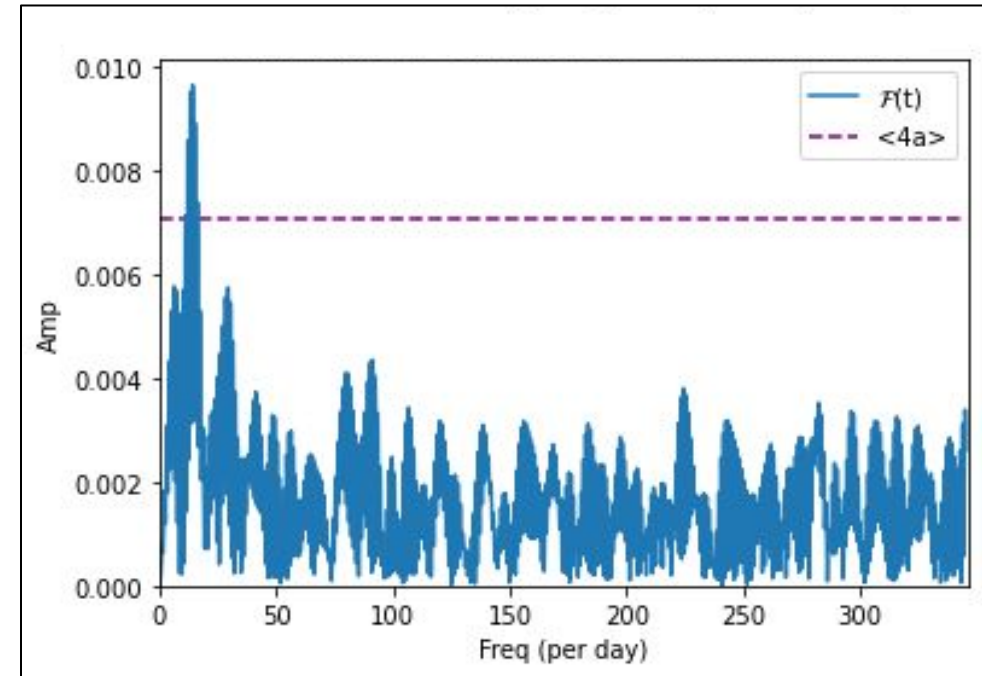


Bootstrapped Periodograms

UT July 2nd-4th, 2024



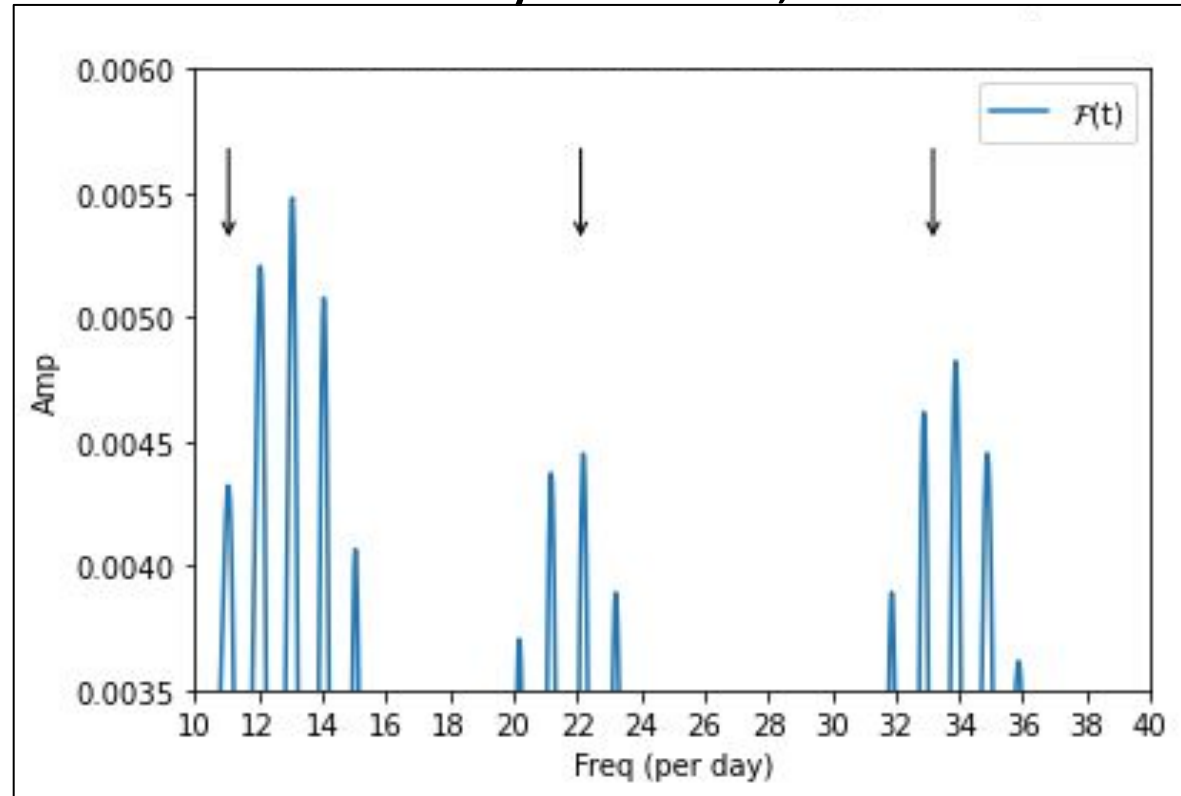
UT April 11th-12th, 2025



Zoomed in Periodogram, July

- Aliases
 - 11.05 /day
 - 12.05 /day
 - 13.05 /day
 - 14.05 /day
- Harmonics
 - Second ≈ 22 /day
 - Third ≈ 33 /day

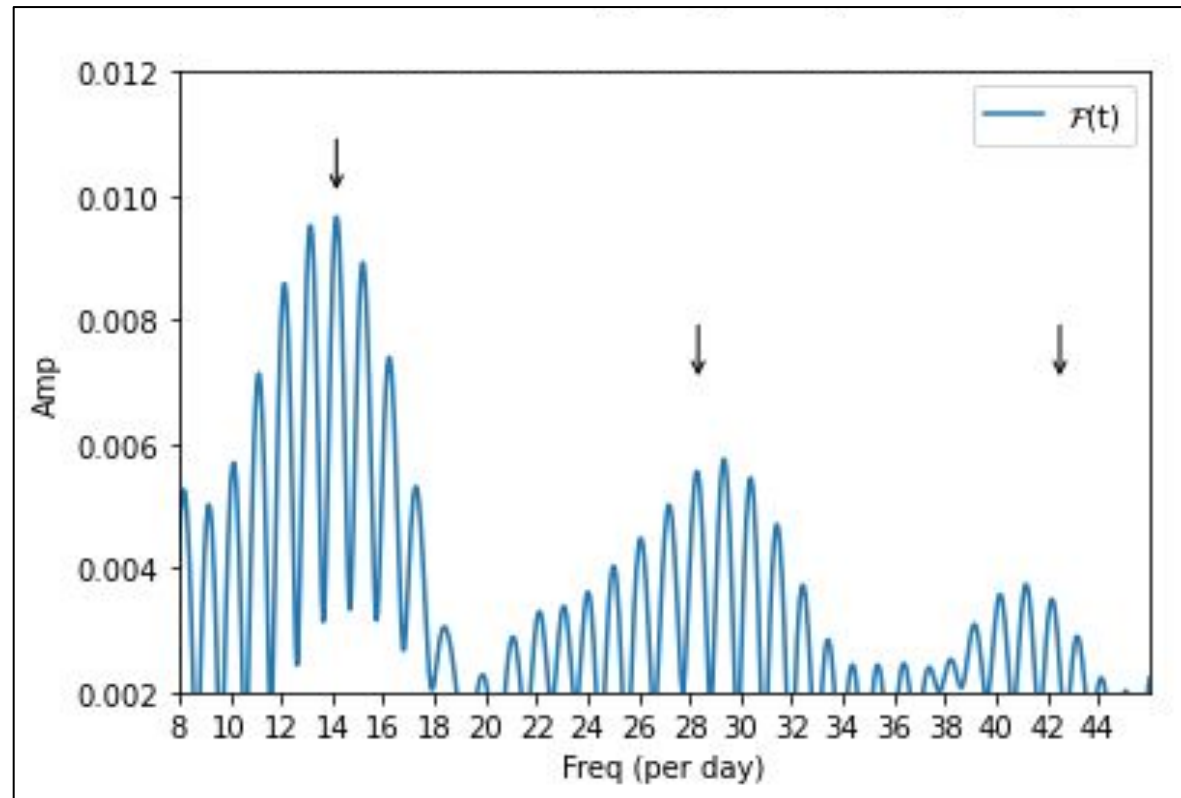
UT July 2nd-4th, 2024



Zoomed in Periodogram, April

- Aliases
 - 11.11 /day
 - 12.16 /day
 - 13.18 /day
 - 14.18 /day
- Harmonics
 - Second ≈ 28 /day
 - Third ≈ 42 /day

UT April 11th-12th, 2025



What does this all mean?

- Aliases

- 11.05 /day
- 12.05 /day
- 13.05 /day
- 14.05 /day

- Harmonics

- Second ≈ 22 /day
- Third ≈ 33 /day

- Aliases

- 11.11 /day
- 12.16 /day
- 13.18 /day
- 14.18 /day

- Harmonics

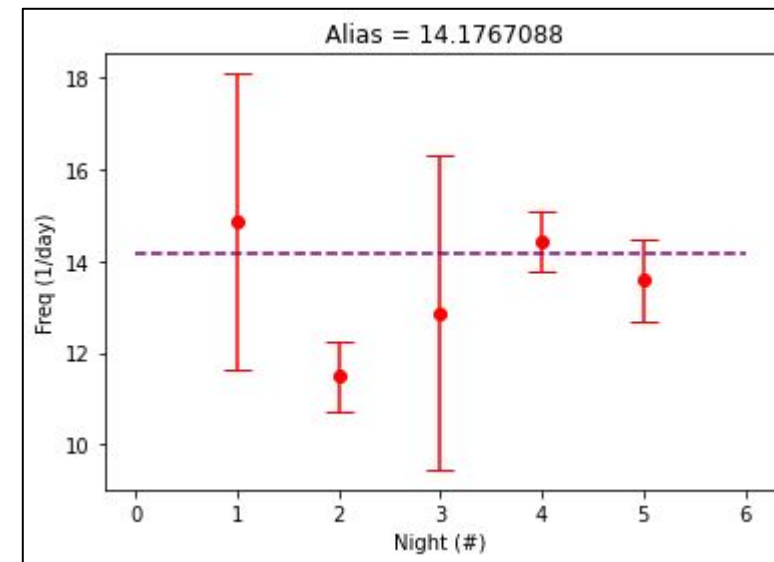
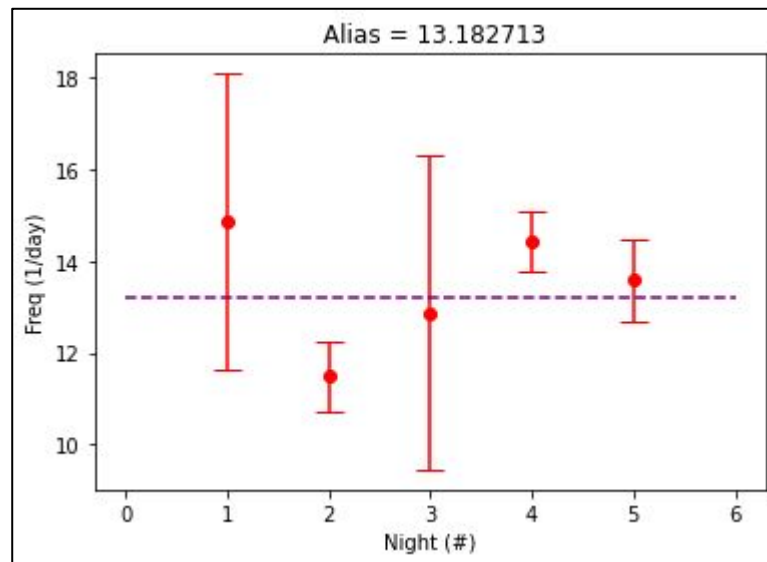
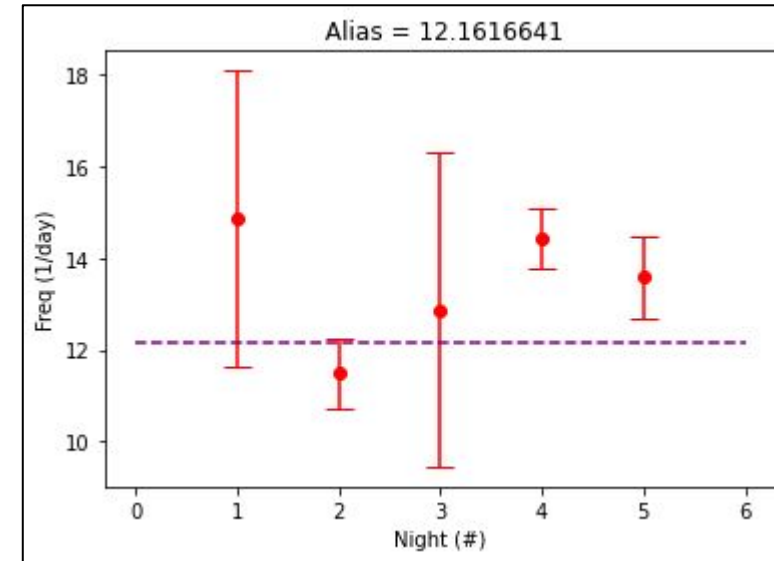
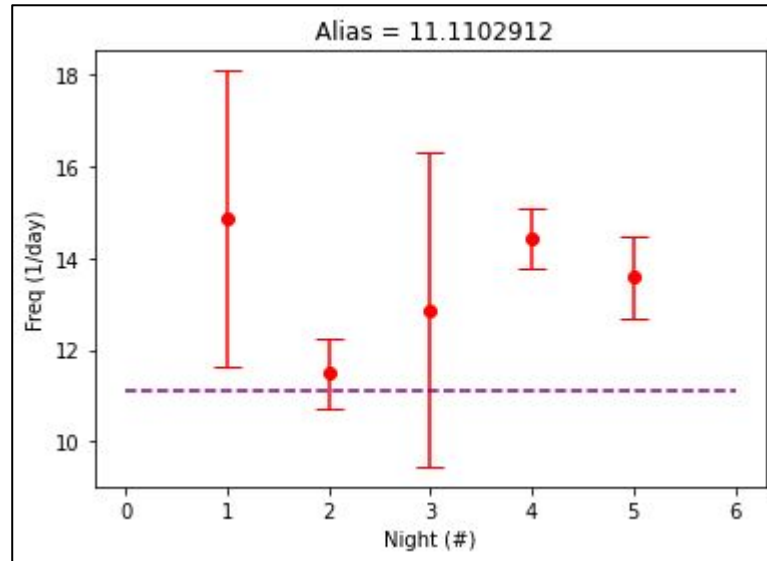
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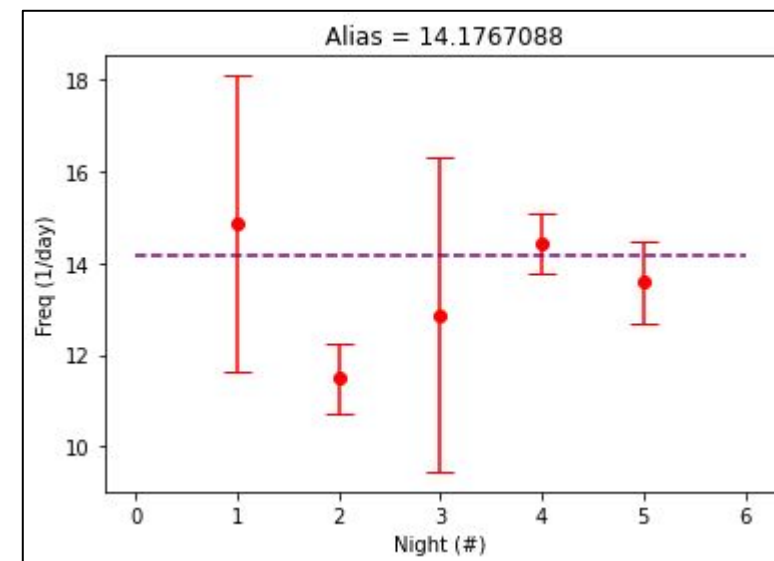
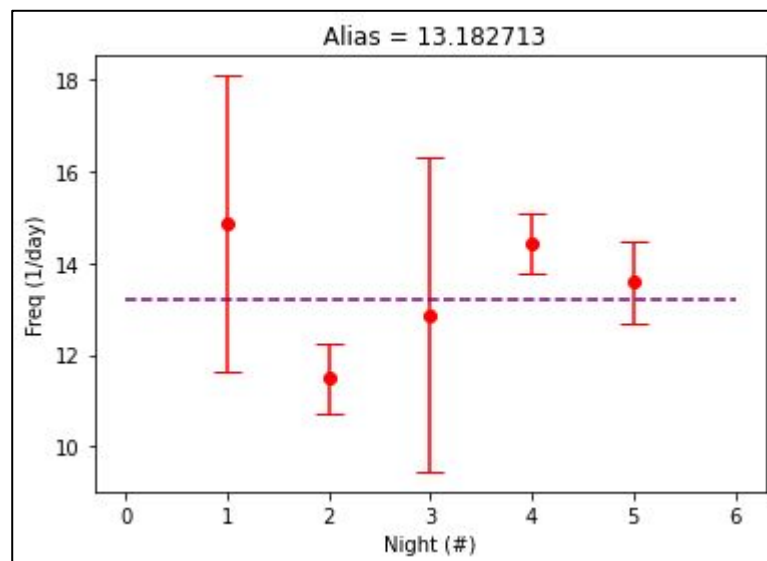
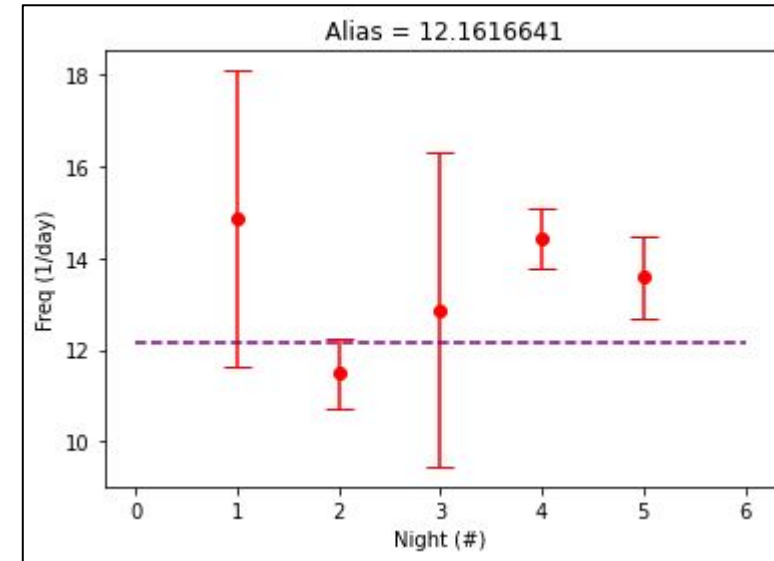
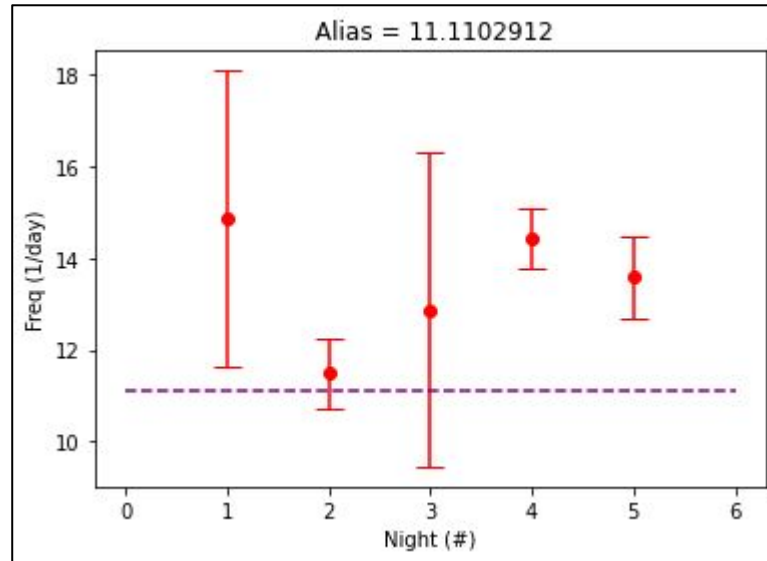
Source Code Used: Doctor Matt Wood



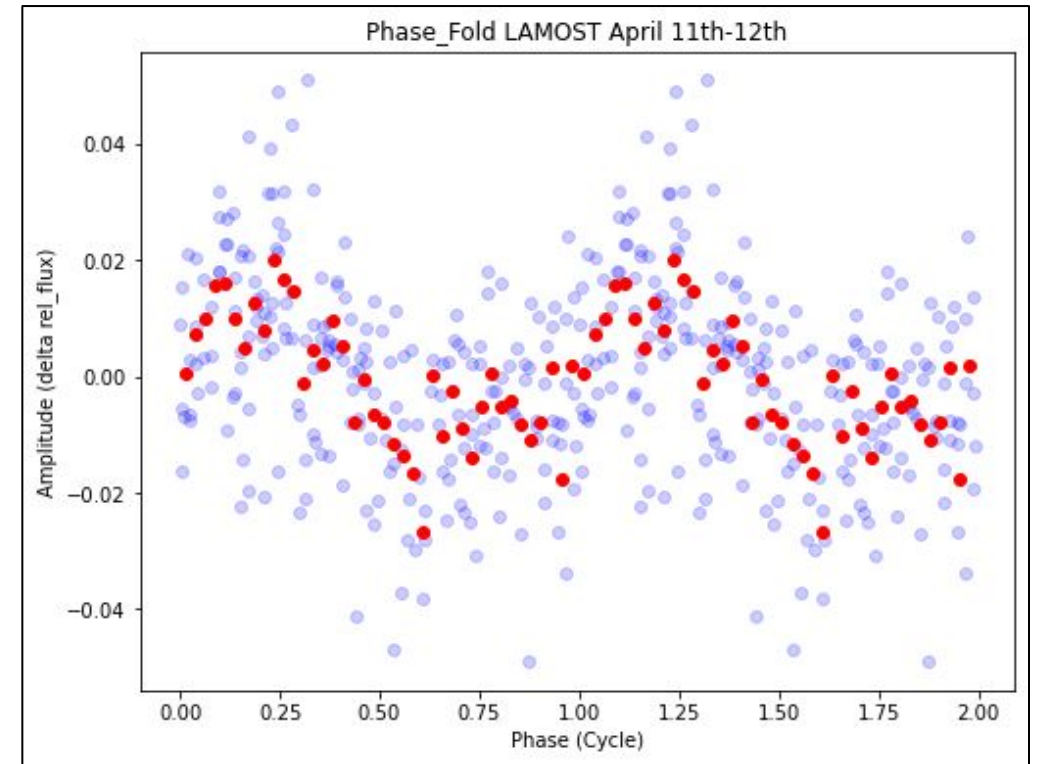
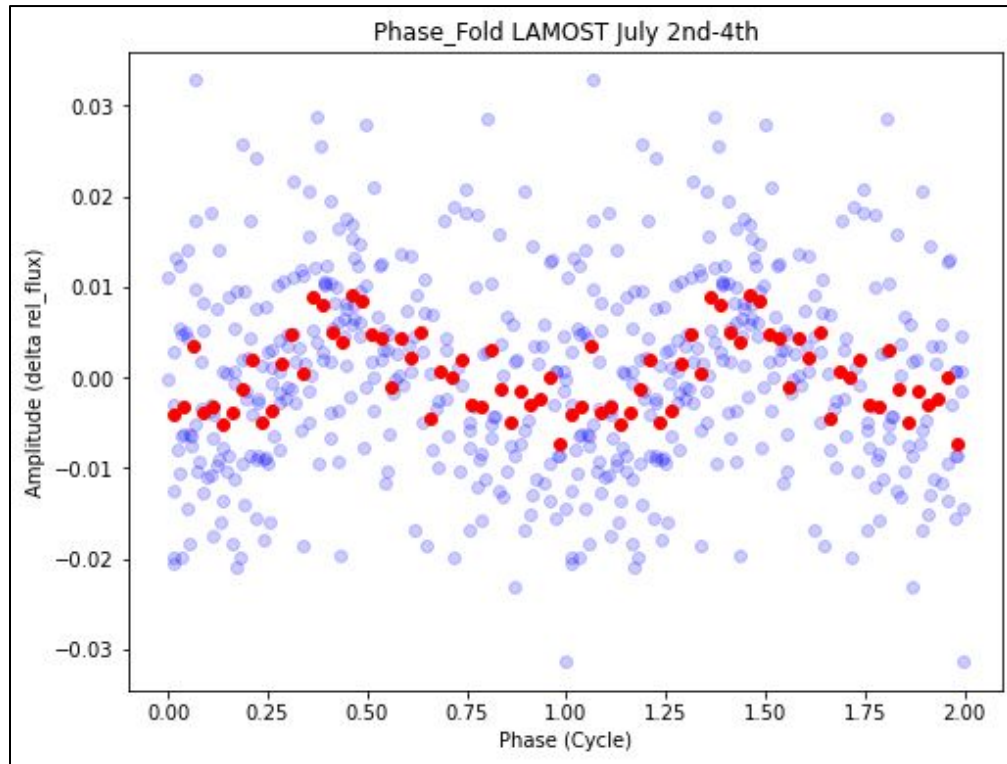
Qualitative Analysis



Qualitative Analysis



Phase Fold Diagrams, Freq ≈ 13 /day



Source Code Used: Doctor Matt Wood



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Results

- The data strongly suggests that LAMOST J1327 is rapidly rotating
 - Freq = 13.18271 ± 0.0503 /day
 - Period = $1.82056607 \pm 0.006955153$ hours
- Possible Future Work
 - Better observations made by TESS
 - Observations of more targets
 - The results from this project can be used to help determine the percentage of rapidly rotating massive WDs due to Double Degenerate WD Mergers



Acknowledgments

- NSF - Funding
- ETAMU PHAS Department for hosting the REU
- Doctor Kurtis Williams for advising me
- My Peers

